

The LDO was designed with a PMOS pass-device topology to minimize dropout voltage and avoid the voltage headroom limitations inherent in an NMOS source-follower. The output voltage is regulated to $V_{OUT} = 1.2V$ using a reference voltage of $V_{REF} = 0.8V$, yielding a feedback factor of $\beta \approx 0.67$ ($R_1 = 0.5R_2$). To minimize quiescent current, a larger value was chose, but not so large that it compromised stability ($R_1 = 100k\Omega$, $R_2 = 200k\Omega$). To meet the required load and line regulation specifications, the Error Amplifier (EA) was designed to provide a DC loop gain of approximately 75dB, as governed by the following approximations:

$$\text{Load Reg: } \frac{V_{OUT}}{V_{REF}} \approx \frac{1}{\beta} \quad \text{if } A_{EA0} \cdot g_{mp} \cdot R_{FL} \cdot \beta \gg 1 \quad (1)$$

$$\text{Line Reg: } \frac{V_{OUT}}{V_{IN}} \approx \frac{1}{A_{EA0} \cdot \beta} \quad \text{if } g_{mp} r_{dsp} \gg 1 \quad (2)$$

Stability is maintained across the full load range (0.1 mA to 10 mA) through a Miller compensation network. A 200 fF compensation capacitor is used to achieve pole-splitting, ensuring the pole at the output of the error amplifier remains dominant. To prevent the Right-Half-Plane zero from degrading the phase margin, a 1.2 k Ω zero-nulling resistor (R_Z) was placed in series with C_C . This resistor shifts the zero into the Left-Half-Plane, providing significant phase lead that stabilizes the system. While the Miller capacitor introduces a path for supply noise coupling at high frequencies, the high DC gain ensures excellent low-frequency PSRR and line regulation.

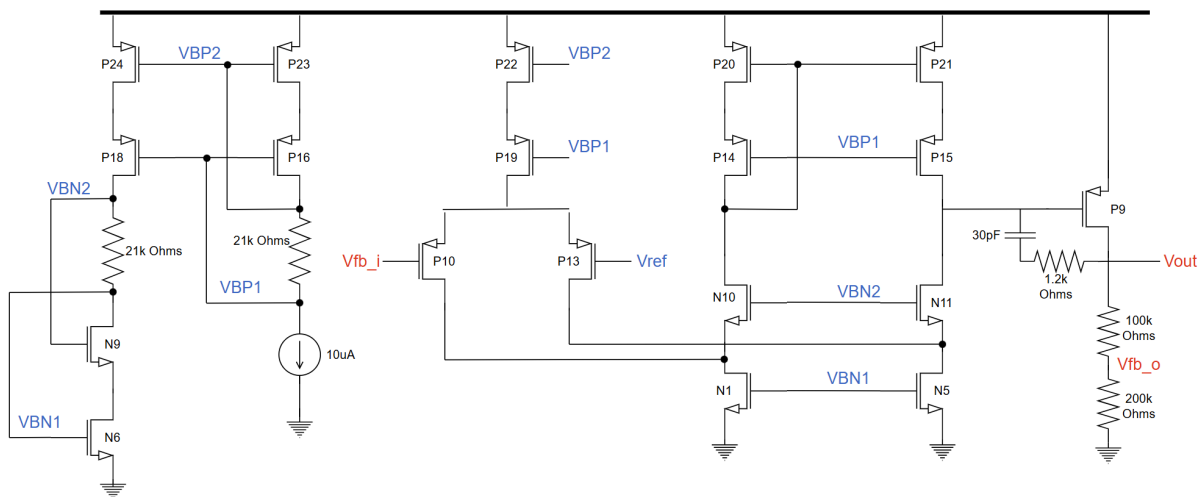


Figure 1: LDO schematic.

Table 1: Transistor Operating Point Summary (Magnitudes)

Device	Size (W/L)	I_D	g_m	Δ
P10/P13	43.2 μm /270 nm	24.87 μA	495.00 μS	36.60 mV
P14/P15	21.6 μm /810 nm	5.37 μA	97.50 μS	71.80 mV
P16/P18	21.6 μm /810 nm	$\approx 10 \mu\text{A}$	$\approx 154 \mu\text{S}$	$\approx 107 \text{ mV}$
P19/P22	36.0 μm /270 nm	49.74 μA	695 μS	128 mV
P20/P21	21.6 μm /540 nm	5.37 μA	105.50 μS	49.50 mV
P23/P24	21.6 μm /540 nm	10.01 μA	118.80 μS	160.30 mV
P9 (Pass)	1000 μm /180 nm	1.00 mA	19.43 mS	43.70 mV
N1/N5	8.4 μm /1.0 μm	30.24 μA	336.70 μS	181.00 mV
N6	2.8 μm /1.0 μm	10.01 μA	111.00 μS	180.80 mV
N9	8.4 μm /1.0 μm	10.01 μA	170.70 μS	94.10 mV
N10/N11	8.4 μm /1.0 μm	5.37 μA	106.30 μS	57.80 mV

Table 2: Performance Summary

Design Parameter/Variable	Simulated Performance	Specification
Input Voltage	1.8 V $\pm$ 10%	1.8 V $\pm$ 10%
Output Voltage	1.2 V $\pm$ 0.0037%	1.2 V $\pm$ 0.5%
Load Current	0.1 mA – 10 mA	0.1 mA – 10 mA
DC Load Regulation	4.83 $\mu\text{V}/\text{mA}$	$\leq 50 \mu\text{V}/\text{mA}$
DC Line Regulation	5.87 $\mu\text{V}/\text{V}@0.1 \text{ mA}$ 11.13 $\mu\text{V}/\text{V}@10 \text{ mA}$	$\leq 500 \mu\text{V}/\text{V}$
Quiescent Current ($I_L = 0.1/10 \text{ mA}$)	84.5 μA /84.49 μA	Minimum
PSR ($@F_{in} = 1 \text{ kHz}/1 \text{ MHz}$)	–63.85 dB@1 kHz, 0.1 mA –3.74 dB@1 MHz, 0.1 mA –63.84 dB@1 kHz, 10 mA –6.27 dB@1 MHz, 10 mA	—
Worst-case PSR ($I_L = 0.1/10 \text{ mA}$)	–1.172 dB/ –0.4605 dB	—
DC Loop Gain ($I_L = 0.1/10 \text{ mA}$)	94.75 dB/90.53 dB	—
Unity Gain Frequency ($I_L = 0.1/10 \text{ mA}$)	1.25 MHz/1.46 MHz	—
Phase Margin ($I_L = 0.1/10 \text{ mA}$)	61.1°/100.2°	$\geq 45^\circ / \geq 60^\circ$
Gain Margin ($I_L = 0.1/10 \text{ mA}$)	32.73 dB/23.91 dB	—

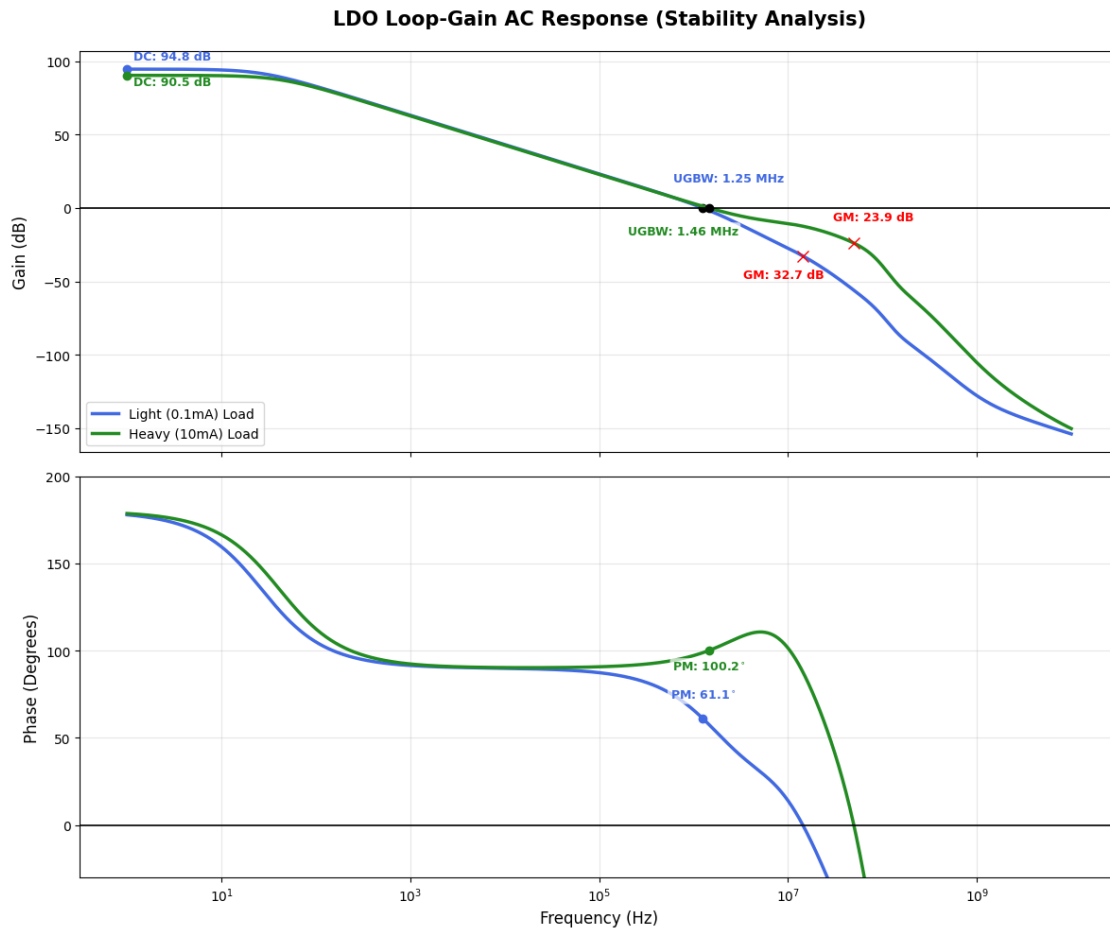


Figure 2: Loop AC Response (Magnitude and Phase)

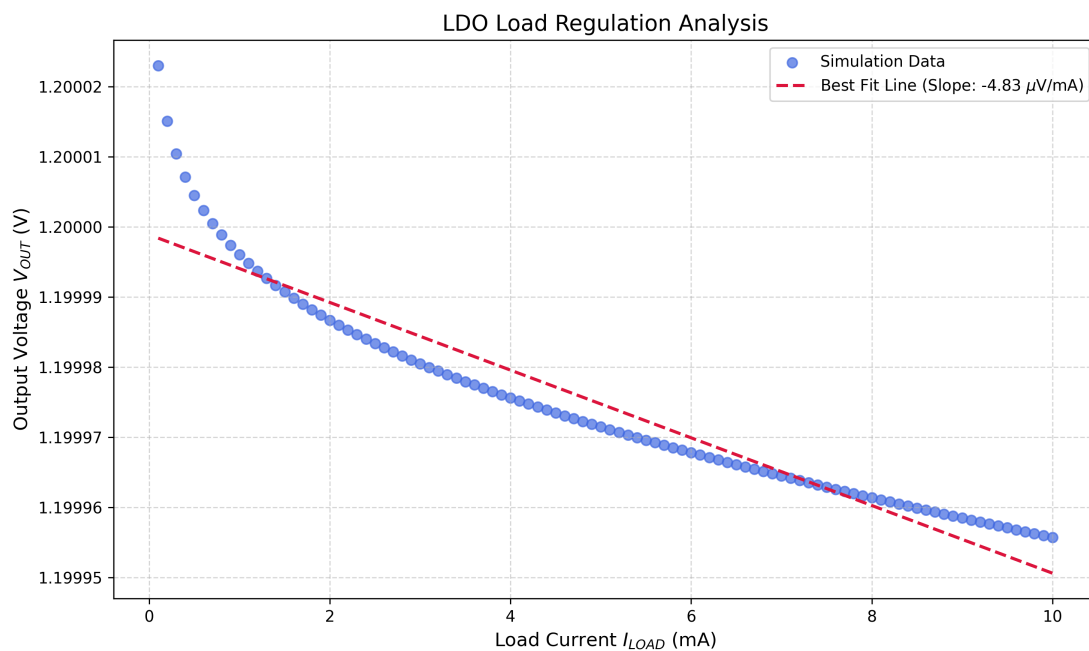


Figure 3: Load Regulation: Output Voltage vs. Load Current

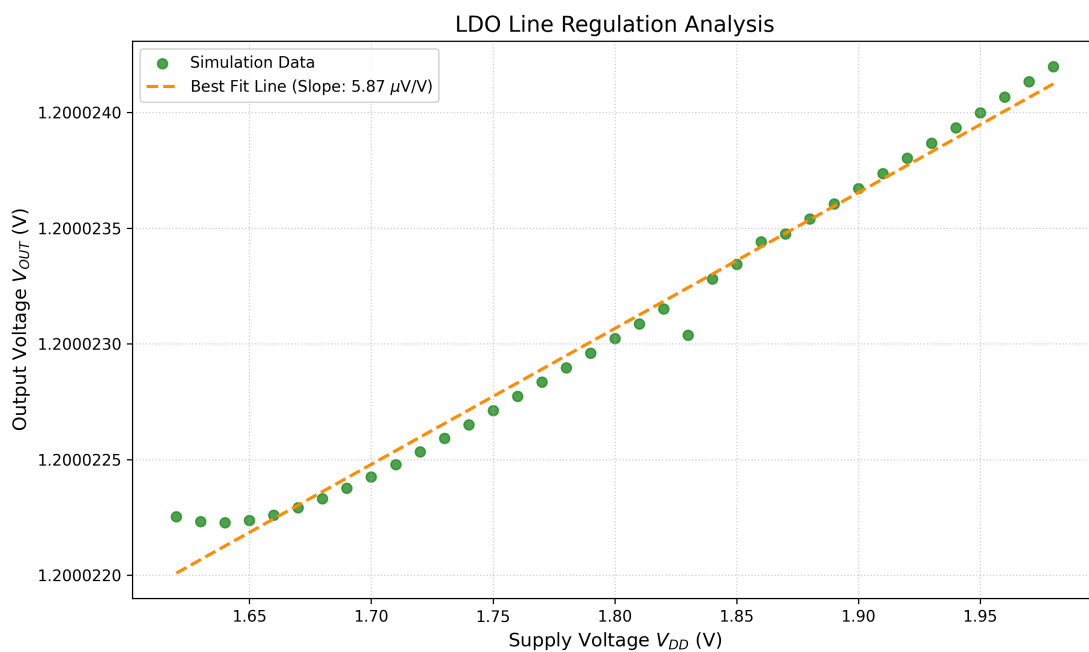


Figure 4: Line Regulation at 0.1 mA Load

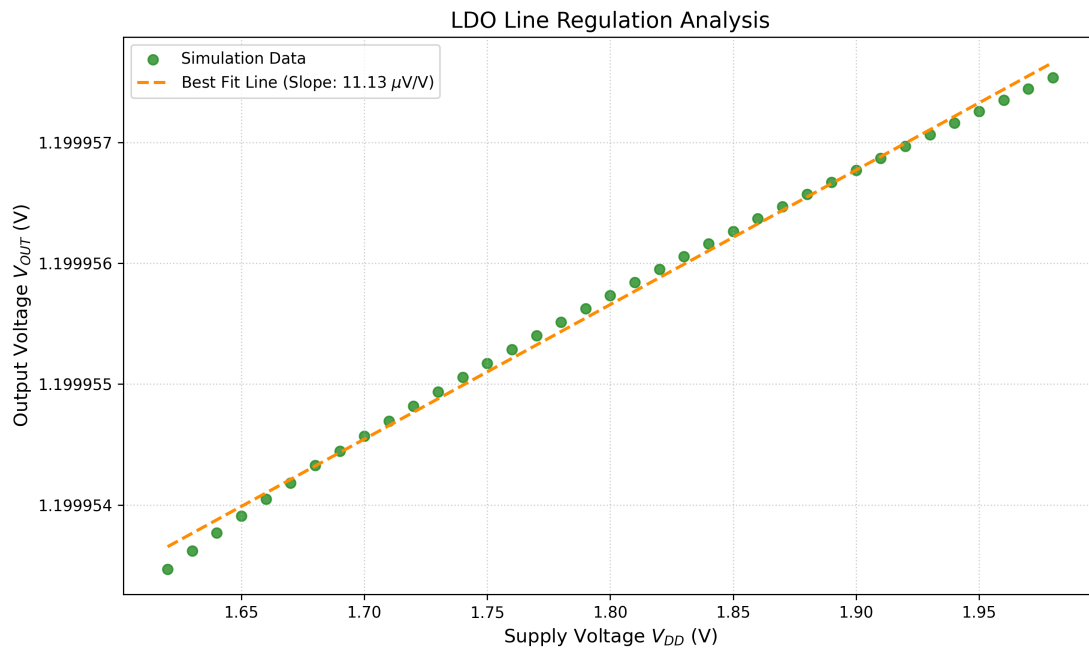


Figure 5: Line Regulation at 10 mA Load

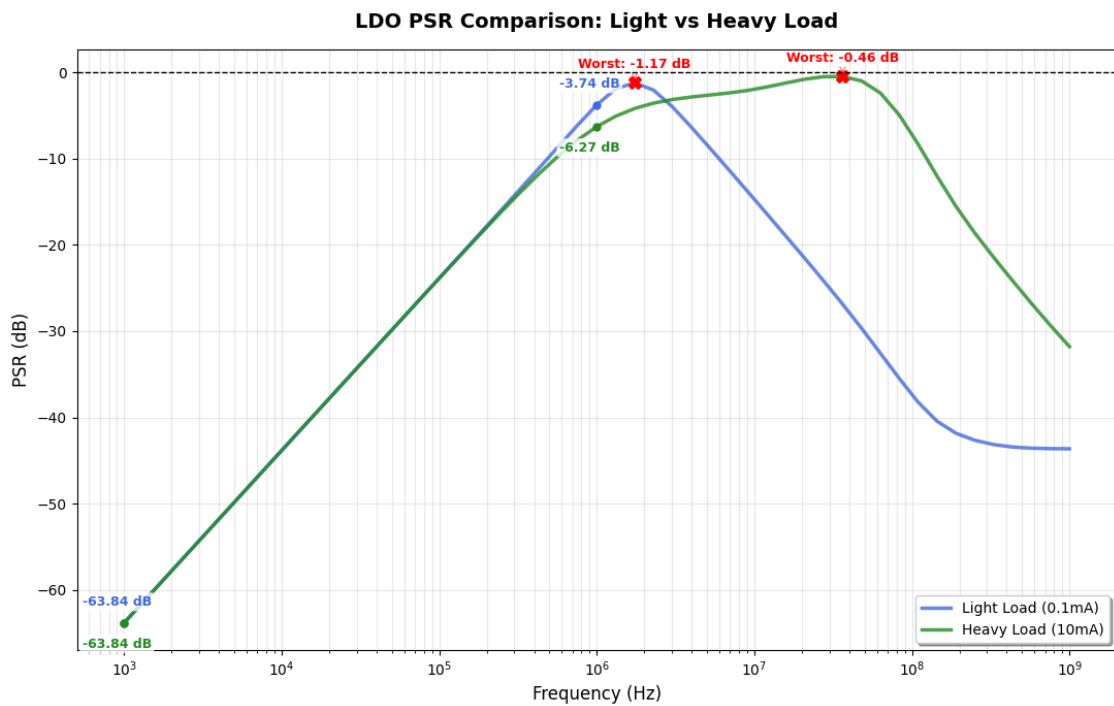


Figure 6: Power Supply Rejection Ratio (PSRR) across Frequency